

When Voltage-Safe Grids Fail: Hidden Dynamic Margins, Weak Nodes, and Hosting Limits in Low-Inertia Power Networks

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Abstract—This paper introduces a unified graph-spectral framework for diagnosing, quantifying, and mitigating vulnerability in low-inertia power networks. Starting from Kirchhoff-consistent linearized power flow and heterogeneous second-order nodal dynamics, we derive closed-form indicators for nodal weakness, dynamic hosting capacity, hidden stability margins, and optimal allocation of inertia, damping, and transmission reinforcements. All results are computable without nonlinear time-domain simulation and reveal fundamental limitations of voltage-only security criteria.

Index Terms—Low-inertia grids, Laplacian dynamics, spectral graph theory, dynamic hosting capacity, hidden stability margins, inverter-based resources.

I. INTRODUCTION

- A. *Motivation: Why Voltage-Safe Grids Still Fail Dynamically*
- B. *Related Work and Limitations of Existing Approaches*

II. SYSTEM MODEL

- A. *Scope and Modeling Assumptions*

We consider a structure-preserving, nodal formulation for the analysis of modern AC power systems operating under low-inertia conditions. The model retains the exact nonlinear AC power-flow relations and bus-level electromechanical dynamics, while explicitly accounting for network-mediated interactions induced by the electrical interconnection structure. All electrical couplings are derived directly from Kirchhoff's and Ohm's laws and are represented at the level of network buses, allowing heterogeneous resources—including synchronous generators, inverter-based resources (IBRs), and aggregated loads—to be treated within a unified mathematical framework. The scope of the model is intentionally restricted. Electromagnetic transients, switching phenomena, large disturbances such as faults or line outages, and protection or relay dynamics are not considered. The focus is instead placed on electromechanical and power-flow time scales relevant to frequency, angle, and voltage dynamics in inverter-dominated networks. These assumptions enable a mathematically tractable yet physically consistent representation suitable for analytical investigation of network-coupled nonlinear dynamics.

- B. *Electrical Network as a Mathematical Object*

The electrical network is modeled as an undirected and connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where the node set $\mathcal{V} = \{1, \dots, n\}$

corresponds to electrical buses and the edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents transmission lines or transformers. Rather than serving as a purely topological abstraction, the graph $\mathcal{G}$ defines the structural support through which all electrical interactions are mediated.

The physical and electrical properties of the network are fully encoded by the complex bus admittance matrix $Y_{\text{bus}} \in \mathbb{C}^{n \times n}$, expressed as

$$Y_{\text{bus}} = G + jB,$$

where G and B denote the conductance and susceptance matrices, respectively. The admittance matrix defines a linear operator mapping nodal voltages to nodal current injections, providing an exact algebraic realization of Ohm's law at the branch level and Kirchhoff's current law at the nodal level. Its sparsity pattern is induced by the graph $\mathcal{G}$, while its entries encode both network topology and electrical parameters. At each bus $i \in \mathcal{V}$, the electrical state is described by a complex voltage phasor $V_i = |V_i|e^{j\theta_i}$, where $|V_i| > 0$ and $\theta_i \in \mathbb{R}$ denote the voltage magnitude and phase angle, respectively. Collecting all nodal quantities yields the voltage vectors $|V| \in \mathbb{R}_{>0}^n$ and $\theta \in \mathbb{R}^n$, which together parametrize the network operating point. Under this representation, electrical interactions arise as nonlinear, operating-point-dependent couplings structured by the graph $\mathcal{G}$ and induced by the admittance operator Y_{bus} . Formally, the nodal voltages and currents satisfy the algebraic relation in equation (1) which constitutes the foundational network equation underlying all subsequent power-flow and dynamic formulations.

$$I = Y_{\text{bus}}V, \quad (1)$$

- C. *Electrical Variables and Nodal Quantities*

At each bus $i \in \mathcal{V}$, the electrical interaction with the network is characterized by the net complex power injection $S_i = P_i + jQ_i$, where $P_i \in \mathbb{R}$ and $Q_i \in \mathbb{R}$ denote the active and reactive power injections, respectively. These quantities represent the aggregated effect of all devices connected at the corresponding bus, including generation, inverter-based resources, and loads, and are treated as nodal variables without resolving internal device-level structure. Collecting the nodal power injections yields the active and reactive power vectors $P \in \mathbb{R}^n$ and $Q \in \mathbb{R}^n$, which, together with the voltage magnitude and phase angle vectors $|V| \in \mathbb{R}_{>0}^n$ and $\theta \in \mathbb{R}^n$, define the electrical state of the network. At this stage, no temporal

evolution is imposed on these variables; they are introduced purely as algebraic quantities describing a generic operating point of the AC network. The nodal electrical variables are related through the standard complex power definition, which links power injections, voltages, and currents at each bus. This relation enforces instantaneous power balance at the nodal level and serves as the interface between network physics and subsequent dynamic modeling. Formally, for each bus $i \in \mathcal{V}$, the complex power injection satisfies (2) where V_i and I_i denote the nodal voltage and current, respectively, and $(\cdot)^*$ denotes complex conjugation.

$$S_i = P_i + jQ_i = V_i I_i^*, \quad (2)$$

D. Network Laws: Kirchhoff and Ohm

The electrical interaction among buses is governed exclusively by fundamental network laws. Specifically, Ohm's law at the branch level and Kirchhoff's current law at the nodal level jointly determine how nodal voltages induce current injections through the network. No additional coupling assumptions are introduced: all electrical interactions are fully mediated by the network admittance operator. Under the nodal formulation adopted in this work, the currents injected at the network buses arise as the linear image of the complex voltage vector through the bus admittance matrix. This relation encodes both the physical parameters of the transmission elements and the interconnection structure defined by the graph $\mathcal{G}$, thereby ensuring that all nodal interactions are physically consistent and network-induced. Formally, the nodal current vector $I \in \mathbb{C}^n$ and the nodal voltage vector $V \in \mathbb{C}^n$ satisfy the algebraic network equation (3) which constitutes the fundamental physical constraint underlying the AC power network.

$$I = Y_{\text{bus}} V, \quad (3)$$

E. Nonlinear AC Power-Flow Relations

The steady-state electrical interactions among buses are obtained by combining the nodal current relation with the complex power definition. Substituting the network law into the nodal power expression yields a set of nonlinear algebraic relations linking voltage magnitudes and phase angles to active and reactive power injections. These relations are exact and follow directly from Kirchhoff's and Ohm's laws, without linearization or simplifying assumptions.

To expose the network-induced structure of the AC power-flow equations, we introduce the phase-difference matrix

$$\Theta(\theta) := \theta \mathbf{1}^\top - \mathbf{1} \theta^\top \in \mathbb{R}^{n \times n},$$

whose (i, j) -th entry equals $\theta_i - \theta_j$. This representation allows the trigonometric coupling between buses to be expressed in a compact, graph-structured form.

Using this notation, the active and reactive electrical power injections induced by the network can be written as nonlinear

functions of the nodal voltages. Specifically, the power-flow relations take the form

$$P_e(\theta, |V|) = \text{diag}(|V|) K_P(\theta) |V|, \quad Q_e(\theta, |V|) = \text{diag}(|V|) K_Q(\theta) |V| \quad (4)$$

where the coupling matrices $K_P(\theta)$ and $K_Q(\theta)$ are defined entrywise by

$$K_P(\theta) = G \circ \cos(\Theta(\theta)) + B \circ \sin(\Theta(\theta)), \quad K_Q(\theta) = G \circ \sin(\Theta(\theta)) - B \circ \cos(\Theta(\theta)) \quad (5)$$

Here, $\circ$ denotes the Hadamard (elementwise) product, and the trigonometric functions act entrywise on $\Theta(\theta)$.

Equations (4)–(5) provide an exact, nonlinear, and Kirchhoff-consistent description of steady-state power exchange in the network. The sparsity patterns of G and B determine the electrical coupling structure, while voltage magnitudes and phase differences modulate interaction strength in an operating-point-dependent manner.

F. Interpretation as a Networked Nonlinear System

The AC power-flow relations derived above admit a natural interpretation as a nonlinear networked system. Each bus $i \in \mathcal{V}$ can be viewed as an electrical agent whose local state is parametrized by the pair $(\theta_i, |V_i|)$ and whose interaction with the rest of the network is mediated exclusively through the graph $\mathcal{G}$ and the admittance operator Y_{bus} . No direct pairwise coupling is imposed a priori; all interconnections emerge from the underlying network physics.

From a structural perspective, the nonlinear power injection maps $P_e(\theta, |V|)$ and $Q_e(\theta, |V|)$ define vector-valued operators

$$P_e : \mathbb{R}^n \times \mathbb{R}_{>0}^n \rightarrow \mathbb{R}^n, \quad Q_e : \mathbb{R}^n \times \mathbb{R}_{>0}^n \rightarrow \mathbb{R}^n,$$

whose components depend on global voltage phase differences and magnitudes. These operators are structured by the graph $\mathcal{G}$ through the sparsity patterns of the conductance and susceptance matrices and induce a nonlinear, dense coupling among nodal variables.

Equivalently, the AC power-flow relations can be regarded as the action of a graph-structured nonlinear operator on the nodal voltage state. In this view, the matrices $K_P(\theta)$ and $K_Q(\theta)$ act as operating-point-dependent coupling operators, whose entries encode both electrical parameters and relative phase information. The dependence on $\Theta(\theta)$ introduces a tensorial structure in which pairwise phase differences are lifted to a matrix-valued object prior to being projected back onto nodal power injections.

This interpretation emphasizes that the electrical network does not merely interconnect otherwise independent subsystems. Instead, it induces a global nonlinear interaction law in which each nodal quantity depends on the collective network state. Such a representation provides a natural foundation for the introduction of distributed dynamics, heterogeneous nodal models, and operator-based analysis tools in subsequent sections.

G. Networked Nonlinear Model (AC-DAE)

a) *Graph and network operators.*: Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected, connected graph with $|\mathcal{V}| = n$ buses and $|\mathcal{E}| = m$

branches. Fix an arbitrary orientation of the edges and let $\mathcal{A} \in \mathbb{R}^{n \times m}$ be the oriented incidence matrix. For each branch $\ell = (i, j) \in \mathcal{E}$, let $y_\ell = g_\ell + j b_\ell$ denote its series admittance, and define the diagonal edge-weight matrices $G_e := \text{diag}(g_\ell)$ and $B_e := \text{diag}(b_\ell)$. The associated weighted (graph) Laplacians are

$$L_G := \mathcal{A} G_e \mathcal{A}^\top, \quad L_B := \mathcal{A} B_e \mathcal{A}^\top. \quad (6)$$

In the general lossy AC setting, the exact network physics are encoded by the bus-admittance matrix $Y_{\text{bus}} = G + jB$ (possibly including shunts and transformers), which induces the nodal current relation $I = Y_{\text{bus}} V$.

b) Electrical variables.: At each bus $i \in \mathcal{V}$, the voltage phasor is $V_i = |V_i| e^{j\theta_i}$ and the net complex power injection is $S_i = P_i + jQ_i$. Stacking nodal quantities yields $\theta \in \mathbb{R}^n$, $|V| \in \mathbb{R}_{>0}^n$, $P, Q \in \mathbb{R}^n$ and $V \in \mathbb{C}^n$.

c) Exact nonlinear AC power-flow operator.: Define the phase-difference matrix $\Theta(\theta) := \theta \mathbf{1}^\top - \mathbf{1} \theta^\top$ and the coupling operators

$$K_P(\theta) = G \circ \cos(\Theta(\theta)) + B \circ \sin(\Theta(\theta)), \quad K_Q(\theta) = G \circ \sin(\Theta(\theta)) + B \circ \cos(\Theta(\theta)). \quad (7)$$

The network-induced active and reactive injections are given by

$$P_e(\theta, |V|) = \text{diag}(|V|) K_P(\theta) |V|, \quad Q_e(\theta, |V|) = \text{diag}(|V|) K_Q(\theta) |V|. \quad (8)$$

d) Per-bus dynamics (structure-preserving nodal swing + injection dynamics).: Partition buses into dynamic buses $\mathcal{V}_d$ and algebraic buses $\mathcal{V}_\ell$ with $\mathcal{V} = \mathcal{V}_d \cup \mathcal{V}_\ell$ and $\mathcal{V}_d \cap \mathcal{V}_\ell = \emptyset$ (functional, not topological). For each $i \in \mathcal{V}_d$, the kinematics and nodal swing dynamics are

$$\dot{\theta}_i = \omega_i, \quad M_i \dot{\omega}_i = -D_i \omega_i + P_i - P_{e,i}(\theta, |V|), \quad (9)$$

where $M_i > 0$ and $D_i \geq 0$ denote aggregated inertia and damping (or frequency-droop-like damping) at bus i .

To capture aggregated IBR and load response at the nodal level, we model active and reactive injections via first-order actuation layers

$$\tau_{P,i} \dot{P}_i = -P_i + P_i^*(|V_i|, \omega_i), \quad \tau_{Q,i} \dot{Q}_i = -Q_i + Q_i^*(|V_i|, \omega_i), \quad (10)$$

with $\tau_{P,i}, \tau_{Q,i} > 0$ and generic static maps $P_i^*(\cdot), Q_i^*(\cdot)$ that represent aggregated internal control or device physics (without resolving device states).

e) Algebraic constraints (AC-DAE closure).: Algebraic buses satisfy nodal power-balance constraints

$$0 = P_\ell - P_{e,\ell}(\theta, |V|), \quad 0 = Q_\ell - Q_{e,\ell}(\theta, |V|), \quad \forall \ell \in \mathcal{V}_\ell, \quad (11)$$

together with standard bus-type constraints (e.g., slack and PV buses). For a slack bus $s \in \mathcal{V}$, one enforces

$$\theta_s = \theta_s^*, \quad |V_s| = |V_s|^*, \quad (12)$$

and for PV buses one typically fixes $|V_i| = |V_i|^*$ while enforcing active-power balance through P_i (with Q_i determined implicitly by the AC power-flow operator).

f) Condensed nonlinear network form.: Let $S_d \in \mathbb{R}^{|\mathcal{V}_d| \times n}$ and $S_\ell \in \mathbb{R}^{|\mathcal{V}_\ell| \times n}$ be selection matrices that extract dynamic and algebraic components, e.g., $\theta_d = S_d \theta$, $\theta_\ell = S_\ell \theta$, and similarly for other vectors. Define diagonal parameter matrices

$$M := \text{diag}(M_i)_{i \in \mathcal{V}_d}, \quad D := \text{diag}(D_i)_{i \in \mathcal{V}_d}, \quad \tau_P := \text{diag}(\tau_{P,i}), \quad \tau_Q := \text{diag}(\tau_{Q,i}),$$

Then the overall AC-DAE model can be written compactly as

$$\dot{\theta}_d = \omega_d, \quad (13a)$$

$$M \dot{\omega}_d = -D \omega_d + P_d - S_d P_e(\theta, |V|), \quad (13b)$$

$$\tau_P \dot{P} = -P + P^*(|V|, \omega), \quad \tau_Q \dot{Q} = -Q + Q^*(|V|, \omega), \quad (13c)$$

$$0 = \begin{bmatrix} S_\ell P - S_\ell P_e(\theta, |V|) \\ S_\ell Q - S_\ell Q_e(\theta, |V|) \end{bmatrix}, \quad \theta_s = \theta_s^*, \quad |V_s| = |V_s|^* \text{ (and PV constraints)} \quad (13d)$$

g) Laplacian viewpoint (graph-structured coupling).:

The operators $P_e(\theta, |V|)$ and $Q_e(\theta, |V|)$ are graph-structured: their coupling is induced by $(\mathcal{G}, Y_{\text{bus}})$. In particular, in the lossless case ($G = 0$) with fixed magnitudes $|V|$, the active-power map reduces to a Laplacian-mediated sine coupling of the form $P_e(\theta, |V|) = \text{diag}(|V|) (B \circ \sin(\Theta(\theta))) |V|$, which is naturally compatible with the weighted susceptance Laplacian $K_Q(\theta)$.

H. Mass-Normalized Laplacian and Damping Operators

Definition 1 (Mass-Normalized Laplacian).

$$S := M^{-1/2} L M^{-1/2}.$$

Definition 2 (Normalized Damping Operator).

$$R := M^{-1/2} D M^{-1/2}.$$

I. Modal Decomposition and Physical Interpretation

III. NODAL WEAKNESS AND VULNERABILITY INDICES

A. Static Stress Indices (Power and Angle)

B. Dynamic Modal Participation and RoCoF Sensitivity

C. Unified Nodal Weakness Score

Definition 3 (Nodal Weakness Index).

$$\mathcal{W}_i = \beta_P S_i^{\text{stat}} + \beta_V S_i^V + \beta_D S_i^{\text{dyn}}.$$

D. Ranking and Interpretation

IV. DYNAMIC HOSTING CAPACITY AND COLLAPSE MARGINS

A. Static Power Transfer Limits

B. Dynamic Frequency and RoCoF Limits

C. Node-Specific and Network-Wide Hosting Capacity

Definition 4 (Dynamic Hosting Capacity).

V. HIDDEN DYNAMIC MARGINS

Definition 5 (Hidden Stability Margin).

$$\text{HM} := \Delta P^{\text{stat}} - \Delta P^{\text{dyn}}.$$

A. *Nodal Hidden Margins*

B. *Interpretation and Failure Scenarios*

Theorem 1 (Voltage Safety Does Not Imply Dynamic Safety).

VI. TRANSMISSION REINFORCEMENT AND EDGE CRITICALITY

A. *Spectral Sensitivity of Network Robustness*

Proposition 1.

$$\frac{\partial \lambda_k}{\partial w_e} = \left(b_e^\top M^{-1/2} u_k \right)^2.$$

B. *Edge Criticality Index*

C. *When Reinforcements Help or Hurt*

VII. OPTIMAL ALLOCATION OF INERTIA AND DAMPING

A. *Problem Formulation with Budget Constraints*

B. *Optimality Conditions and Greedy Laws*

Theorem 2.

C. *Physical Interpretation and Cost-Aware Allocation*

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B. *Regime Transitions and Boundaries*

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B. *Ranking Accuracy and Failure Prediction*

C. *Hidden Margin Demonstrations*

D. *Optimal Reinforcement Results*

X. CONCLUSIONS AND OUTLOOK

A. *Summary of Findings*

B. *Implications for Grid Planning and Operation*

C. *Future Work*

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